

Mathematics for Data Science 2

Linear Algebra & Multivariable Calculus for Data Science

IIT Madras BS Degree Programme

IITM BS Degree Curriculum Team

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Introduction

Welcome to **Mathematics for Data Science 2 (MDS2)**. Linear algebra and multivariable calculus form the core mathematical foundation for machine learning algorithms, deep learning neural networks, optimization routines, and dimensionality reduction techniques like SVD and PCA.

Structure of this Guide

The course is divided into **12 Chapters**:

- **Chapters 1–4 (Quiz 1 Content):** Vector Spaces, Matrix Algebra, Systems of Linear Equations, Gaussian Elimination, Subspaces, Basis, and Dimension.
- **Chapters 5–8 (Quiz 2 Content):** Null Space, Linear Transformations, Rank-Nullity Theorem, Inner Product Spaces, Gram-Schmidt Orthogonalization, and Projections.
- **Chapters 9–12 (End-Term Content):** Multivariable Differential Calculus, Gradients, Tangent Planes, Hessian Matrices, Steepest Descent, Eigenvalues, Eigenvectors, Diagonalization, and Singular Value Decomposition (SVD).

Visual Layout & Box Legend

- **Teal Cards** — Definitions, vector spaces, and terminology.
- **Slate Blue Cards** — Theorems, matrix properties, and mathematical rules.
- **Purple Cards** — Step-by-step solved linear algebra examples.
- **Violet Cards** — Geometric intuition & transformation concepts.
- **Amber Cards** — Key formulas and computational algorithms.
- **Red Cards** — IIT Madras & visual YouTube video resources.

Chapter 1

Vectors, Matrices & Determinants

Welcome to Mathematics for Data Science 2! Linear algebra is the foundational mathematical backbone of modern Machine Learning, Artificial Intelligence, and Data Science. From storing high-dimensional feature spaces to computing neural network transformations and performing dimensionality reduction, vectors and matrices are the fundamental building blocks.

1.1 Vectors in $\mathbb{R}^n$

A **Vector** in n -dimensional real space $\mathbb{R}^n$ is an ordered sequence of n real numbers, written as a column matrix:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

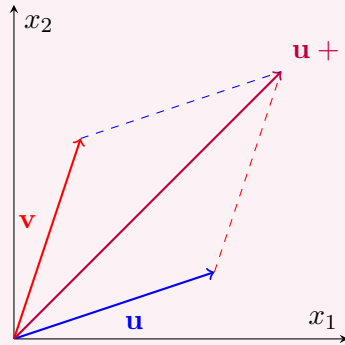
Definition: Vector Operations

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and scalar $c \in \mathbb{R}$:

- **Vector Addition:** $\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$
- **Scalar Multiplication:** $c\mathbf{v} = \begin{bmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{bmatrix}$
- **Dot Product:** $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \sum_{i=1}^n u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$

Core Intuition: Geometric Vector Addition (Parallelogram Law)

While vector addition is just adding numbers row-by-row, geometrically it represents combining two movements in space. If you walk along vector $\mathbf{u}$ and then walk along vector $\mathbf{v}$, your total displacement is $\mathbf{u} + \mathbf{v}$. This forms the diagonal of the parallelogram created by $\mathbf{u}$ and $\mathbf{v}$.



Data Science Relevance: In data science, a sample (e.g. a user profile with age, income, credit score) is represented as a feature vector $\mathbf{x} \in \mathbb{R}^d$. The linear combination $\mathbf{w}^T \mathbf{x} + b$ forms the prediction score in Linear and Logistic Regression!

Video Resource: IITM BS Lecture: Vectors



Official lecture defining vector spaces, geometric interpretations, and algebraic operations in $\mathbb{R}^n$.

1.2 Matrices and Matrix Operations

A **Matrix** $A \in \mathbb{R}^{m \times n}$ is a rectangular array of m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Theorem / Rule: Matrix Multiplication

If $A \in \mathbb{R}^{m \times k}$ and $B \in \mathbb{R}^{k \times n}$, their product $C = AB \in \mathbb{R}^{m \times n}$ has entries:

$$c_{ij} = \sum_{p=1}^k a_{ip} b_{pj}$$

Matrix multiplication is associative $(AB)C = A(BC)$ and distributive, but **NOT commutative** ($AB \neq BA$ in general).

Core Intuition: Matrix Multiplication as Geometric Transformation

Matrix multiplication isn't just an arbitrary set of dot products—it's a geometric transformation of space!

- When you multiply a vector $\mathbf{v}$ by a matrix A ($A\mathbf{v}$), the matrix A grabs the vector and rotates, scales, or shears it to a new location in space.
- When you multiply two matrices together ($C = AB$), you are composing two transformations. First apply transformation B , then apply transformation A .
- Because doing Transformation B then A is usually different than doing A then B , matrix multiplication is **not commutative** ($AB \neq BA$).

Video Resource: IITM BS Lecture: Matrices 

Official lecture explaining matrix algebra, transposition A^T , symmetric matrices ($A = A^T$), and matrix-vector products.

1.3 Systems of Linear Equations

A system of m linear equations in n variables $x_1, x_2, \dots, x_n$ can be compactly written in matrix form:

$$A\mathbf{x} = \mathbf{b}$$

where $A \in \mathbb{R}^{m \times n}$ is the coefficient matrix, $\mathbf{x} \in \mathbb{R}^n$ is the vector of unknowns, and $\mathbf{b} \in \mathbb{R}^m$ is the target output.

Definition: Types of Solutions

Every linear system $A\mathbf{x} = \mathbf{b}$ has either:

1. Exactly **one unique solution** (consistent, non-singular).
2. **Infinitely many solutions** (consistent, dependent).
3. **No solution** (inconsistent, parallel constraints).

Video Resource: IITM BS Lecture: Systems of Linear Equations 

Official lecture introducing linear systems, augmented matrices $[A \mid \mathbf{b}]$, and geometric interpretations of solutions.

1.4 Determinants

The **Determinant** $\det(A)$ or $|A|$ is a scalar value associated with a square matrix $A \in \mathbb{R}^{n \times n}$ that measures the scaling factor of transformation area/volume.

Key Formula / Algorithm: Determinant Computation (2×2 and 3×3)

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\det(A) = ad - bc$$

For $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, expanding along the first row:

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Invertibility Criterion: Matrix A is invertible (A^{-1} exists) if and only if $\det(A) \neq 0$.

Video Resource: IITM BS Lecture: Determinants (Parts 1 & 2)



Official lectures explaining cofactor expansion, properties of determinants, and invertibility conditions.

1.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Vector Dot Product & Angle:** Given vectors $\mathbf{u} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}^T$:

- (a) Compute the dot product $\mathbf{u} \cdot \mathbf{v}$.
- (b) Compute the norms $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$.

2. **Matrix Multiplication:** Given matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix}$$

Compute AB and BA . Verify that $AB \neq BA$.

3. **Matrix-Vector System Formulation:** Convert the linear system:

$$2x_1 + 3x_2 - x_3 = 7$$

$$4x_1 - x_2 + 2x_3 = 3$$

$$-x_1 + 5x_2 + 3x_3 = 10$$

into the matrix form $A\mathbf{x} = \mathbf{b}$ and write down the augmented matrix $[A \mid \mathbf{b}]$.

4. **Determinant Calculation:** Calculate the determinant of the 3×3 matrix:

$$M = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 4 & 5 \\ 1 & 2 & 1 \end{bmatrix}$$

Is matrix M invertible?

5. **Data Science Application (Covariance Matrix Determinant):** In multivariate statistics, the determinant of a 2×2 Covariance Matrix $\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}$ represents total generalized variance. Given $\sigma_1^2 = 4$, $\sigma_2^2 = 9$, and covariance $\sigma_{12} = 3$, calculate $\det(\Sigma)$.

Chapter 2

Cramer's Rule, Row Echelon Form & Gaussian Elimination

Solving large systems of linear equations $Ax = b$ is the single most common computational operation in numerical algorithms. In this chapter, we master explicit solutions via Cramer's Rule and systematic algorithmic reduction via **Gaussian Elimination** and **Row Echelon Form (REF)**.

2.1 Cramer's Rule

For a square, non-singular linear system $Ax = b$ with $\det(A) \neq 0$, **Cramer's Rule** gives an explicit formula for each variable x_i using determinants.

Theorem / Rule: Cramer's Rule Formula

Let $A_i(b)$ be the matrix obtained by replacing the i -th column of A with the constant vector b . The unique solution for variable x_i is:

$$x_i = \frac{\det(A_i(b))}{\det(A)}, \quad \text{for } i = 1, 2, \dots, n$$

Example: Solving 2×2 System via Cramer's Rule

$$\text{Solve } \begin{cases} 2x + 3y = 8 \\ 5x - y = 3 \end{cases} :$$

$$\det(A) = \begin{vmatrix} 2 & 3 \\ 5 & -1 \end{vmatrix} = -2 - 15 = -17$$

$$\det(A_x(b)) = \begin{vmatrix} 8 & 3 \\ 3 & -1 \end{vmatrix} = -8 - 9 = -17 \implies x = \frac{-17}{-17} = 1$$

$$\det(A_y(b)) = \begin{vmatrix} 2 & 8 \\ 5 & 3 \end{vmatrix} = 6 - 40 = -34 \implies y = \frac{-34}{-17} = 2$$

Video Resource: IITM BS Lecture: Cramer's Rule 

Official lecture deriving Cramer's Rule and applying it to non-singular linear systems.

2.2 Elementary Row Operations & Row Echelon Form (REF)

While Cramer's Rule is theoretically elegant, calculating determinants for large $n \times n$ matrices is computationally intractable ($\mathcal{O}(n!)$ time). Instead, we transform augmented matrices using **Elementary Row Operations (EROs)** without changing the solution set.

Definition: Elementary Row Operations (EROs)

1. **Row Swap** ($R_i \leftrightarrow R_j$): Swap two rows.
2. **Scalar Multiplication** ($R_i \rightarrow cR_i$): Multiply a row by a non-zero scalar $c \neq 0$.
3. **Row Addition** ($R_i \rightarrow R_i + cR_j$): Add a scalar multiple of row j to row i .

Definition: Row Echelon Form (REF) & Reduced REF (RREF)

A matrix is in **Row Echelon Form (REF)** if:

1. All zero rows are at the bottom of the matrix.
2. The leading non-zero entry (pivot) of a non-zero row is strictly to the right of the pivot in the row above it.
3. All entries below a pivot are zero.

It is in **Reduced Row Echelon Form (RREF)** if, additionally, every pivot entry is 1, and all entries above each pivot are zero.

Video Resource: IITM BS Lecture: Row Echelon Form



Official lecture defining pivot entries, REF conditions, and RREF canonical form.

Video Resource: IITM BS Lecture: Row Reduction Techniques



Official lecture performing row operations step-by-step on rectangular matrices.

2.3 Gaussian Elimination & Back-Substitution

Gaussian Elimination reduces an augmented matrix $[A \mid \mathbf{b}]$ into REF using EROs, followed by **Back-Substitution** to solve for variables.

Key Formula / Algorithm: Gaussian Elimination Algorithm ($\mathcal{O}(n^3)$)

1. **Forward Elimination:** Use EROs to create zeros below each pivot from left to right, producing REF.
2. **Back-Substitution:** Starting from the bottom row, solve for each basic variable and substitute upwards.

Data Science Relevance: Gaussian Elimination and LU Decomposition under grid computing power the solver engines of NumPy ('np.linalg.solve'), SciPy, and Linear Programming solvers.

Example: Solving Linear Systems in Python

While Gaussian Elimination is vital to understand theoretically, Data Scientists never perform Row Echelon reductions by hand. We use optimized BLAS/LAPACK libraries via NumPy:

```
1 import numpy as np
2
3 # Solve the system:
4 # 2x + 3y = 8
5 # 5x - y = 3
6 A = np.array([[2, 3],
7               [5, -1]])
8 b = np.array([8, 3])
9
10 # np.linalg.solve uses highly optimized LU decomposition (which is just
11 # Gaussian elimination)
12 x = np.linalg.solve(A, b)
13 print(f"Solution: x = {x[0]:.1f}, y = {x[1]:.1f}") # Solution: x = 1.0, y = 2.0
```

Video Resource: IITM BS Lecture: The Gaussian Elimination Method 

Official lecture executing full Gaussian elimination and identifying free vs pivot variables.

2.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Cramer's Rule (2×2):** Use Cramer's Rule to solve the system:

$$\begin{aligned} 3x_1 + 4x_2 &= 10 \\ x_1 - 2x_2 &= -3 \end{aligned}$$

2. **Identifying REF:** State whether each of the following matrices is in REF, RREF, or neither:

$$(a) A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}, \quad (b) B = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}, \quad (c) C = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \end{bmatrix}$$

3. **Gaussian Elimination (3×3):** Solve the linear system using Gaussian Elimination (forward elimination to REF + back-substitution):

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 8 \\ 2x_1 + 3x_2 + 4x_3 &= 20 \\ 4x_1 + 3x_2 + 2x_3 &= 16 \end{aligned}$$

4. **Free Variables & Infinite Solutions:** Reduce the augmented matrix to REF and write the general parametric solution in terms of free variables:

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 4 \\ 2x_1 + 4x_2 + x_3 &= 11 \end{aligned}$$

5. **Data Science Application (Least Squares Normal Equations):** In linear regression, the parameters $\mathbf{w}$ are found by solving $(X^T X)\mathbf{w} = X^T \mathbf{y}$. Given:

$$X^T X = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}, \quad X^T \mathbf{y} = \begin{bmatrix} 10 \\ 9 \end{bmatrix}$$

Solve for parameter vector $\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$ using Gaussian Elimination or Cramer's Rule.

Chapter 3

Vector Spaces, Subspaces & Linear Independence

Abstract algebra generalizes $\mathbb{R}^n$ into the formal concept of a **Vector Space**. Understanding vector spaces and linear independence is crucial for recognizing feature redundancies and understanding high-dimensional data representations.

3.1 Definition of a Vector Space

Definition: Vector Space Axioms

A **Vector Space** $(V, +, \cdot)$ over a field $\mathbb{R}$ is a set V equipped with vector addition $+$ and scalar multiplication $\cdot$ satisfying 8 fundamental axioms:

1. **Associativity of Addition:** $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
2. **Commutativity of Addition:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
3. **Additive Identity:** There exists $\mathbf{0} \in V$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$.
4. **Additive Inverse:** For every $\mathbf{v} \in V$, there exists $-\mathbf{v} \in V$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
5. **Distributivity I:** $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
6. **Distributivity II:** $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$.
7. **Scalar Associativity:** $c(d\mathbf{v}) = (cd)\mathbf{v}$.
8. **Scalar Identity:** $1 \cdot \mathbf{v} = \mathbf{v}$.

Video Resource: IITM BS Lecture: Introduction to Vector Spaces



Official lecture introducing vector space axioms and canonical examples ($\mathbb{R}^n$, matrix spaces $\mathbb{R}^{m \times n}$, polynomial spaces $\mathbb{P}_n$).

3.2 Vector Subspaces

A subset $W \subseteq V$ of a vector space V is a **Subspace** if W is itself a valid vector space under the same operations.

Theorem / Rule: Subspace Test

A non-empty subset $W \subseteq V$ is a subspace of V if and only if it satisfies **3 closure properties**:

1. **Zero Vector:** $\mathbf{0} \in W$.
2. **Closure under Addition:** If $\mathbf{u}, \mathbf{v} \in W$, then $\mathbf{u} + \mathbf{v} \in W$.
3. **Closure under Scalar Multiplication:** If $\mathbf{v} \in W$ and $c \in \mathbb{R}$, then $c\mathbf{v} \in W$.

Video Resource: IITM BS Lecture: Properties of Vector Spaces & Subspaces 

Official lecture proving subspace criteria and geometric examples (lines/planes through the origin in $\mathbb{R}^3$).

3.3 Linear Combinations, Span, and Linear Independence

Definition: Linear Span and Independence

- **Linear Combination:** $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$.
- **Span:** $\text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\})$ is the set of all possible linear combinations of the vectors.
- **Linear Independence:** A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is **Linearly Independent** if the equation:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

has **ONLY the trivial solution** $c_1 = c_2 = \cdots = c_k = 0$.

If a non-trivial solution exists (at least one $c_i \neq 0$), the vectors are **Linearly Dependent**.

Core Intuition: Degrees of Freedom & Redundancy

Think of vectors as giving you directions to move in space.

- A **Linearly Independent** vector unlocks a brand new dimension. If you have two independent vectors in 3D space, you can travel anywhere on a flat 2D plane. A third independent vector would allow you to step off that plane and explore the entire 3D space.
- A **Linearly Dependent** vector is completely redundant. It's like being given a third direction that just points somewhere on the 2D plane you can already reach using the first two vectors. It provides zero new "degrees of freedom".

Data Science Relevance: In Machine Learning, **multicollinearity** occurs when input feature vectors are linearly dependent (or nearly dependent). This causes severe instability in linear models and matrix inversion $(X^T X)^{-1}$.

Video Resource: IITM BS Lecture: Linear Dependence 

Official lecture defining linear dependence and redundant vector identification.

Video Resource: IITM BS Lecture: Linear Independence (Parts 1 & 2) 

Official lectures testing linear independence using matrix rank and determinant of coefficient matrices.

3.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Subspace Test:** Determine whether $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \mid 2x - 3y = 0 \right\}$ is a valid subspace of $\mathbb{R}^2$. Prove using the 3 subspace criteria.
2. **Non-Subspace Counterexample:** Consider $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \mid x + y = 1 \right\}$. Show why W fails to be a vector subspace of $\mathbb{R}^2$.
3. **Testing Linear Independence (3×3):** Test whether the set of vectors:

$$\left\{ \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 5 \\ 6 \\ 0 \end{bmatrix} \right\}$$

is linearly independent in $\mathbb{R}^3$ by computing the determinant of the matrix formed by placing them as columns.

4. **Linear Dependence & Redundancy:** Find the scalar value k such that the vectors $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ k \end{bmatrix}$ are linearly dependent.
5. **Data Science Application (Feature Multicollinearity):** A data scientist collects three features for housing price prediction: x_1 = size in sq ft, x_2 = size in sq meters, x_3 = number of bedrooms. Since $1 \text{ sq ft} = 0.092903 \text{ sq meters}$, show mathematically that $\{x_1, x_2, x_3\}$ is a linearly dependent set of feature vectors, and explain why one must be removed before regression fitting.

Chapter 4

Basis, Dimension & Matrix Rank

A **Basis** is a minimal generating set of linearly independent vectors that spans a vector space. The number of vectors in any basis defines the unique **Dimension** of the vector space, while **Matrix Rank** counts the dimension of its row and column spaces.

4.1 Basis for a Vector Space

Definition: Basis Definition

A set of vectors $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subset V$ is a **Basis** for a vector space V if it satisfies two conditions:

1. $\mathcal{B}$ is **Linearly Independent**.
2. $\mathcal{B}$ **Spans** V , i.e., $\text{span}(\mathcal{B}) = V$.

Every vector $\mathbf{x} \in V$ can be expressed as a **unique** linear combination of the basis vectors.

Example: Standard Basis for $\mathbb{R}^n$

The standard basis for $\mathbb{R}^3$ is $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$. Any vector $[x, y, z]^T = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$.

Video Resource: IITM BS Lecture: What is a Basis for a Vector Space? 

Official lecture introducing basis criteria and unique coordinate representations.

Video Resource: IITM BS Lecture: Finding Bases for Vector Spaces 

Official lecture demonstrating methods to extract bases from spanning sets.

4.2 Dimension of a Vector Space

Theorem / Rule: Dimension Theorem

All bases for a given finite-dimensional vector space V contain the **exact same number of vectors**. This invariant number is called the **Dimension** of V , denoted $\dim(V)$.

- $\dim(\mathbb{R}^n) = n$.
- $\dim(\mathbb{R}^{m \times n}) = m \cdot n$.
- The zero subspace $\{0\}$ has dimension 0.

Video Resource: IITM BS Lecture: Rank and Dimension for Vector Spaces 

Official lecture defining dimension, subspace dimension bounds ($\dim(W) \leq \dim(V)$), and basis extensions.

4.3 Matrix Rank via Gaussian Elimination

Key Formula / Algorithm: Column Rank, Row Rank, and Matrix Rank

For any matrix $A \in \mathbb{R}^{m \times n}$:

- **Column Rank:** The dimension of the column space $\text{span}(\text{columns of } A)$.
- **Row Rank:** The dimension of the row space $\text{span}(\text{rows of } A)$.
- **Fundamental Theorem:** $\text{Row Rank}(A) = \text{Column Rank}(A) = \text{rank}(A)$.
- **REF Calculation:** $\text{rank}(A)$ is equal to the **number of non-zero rows (or pivot columns)** in the Row Echelon Form (REF) of A .

Data Science Relevance: In **Principal Component Analysis (PCA)** and low-rank matrix approximations (SVD), matrix rank tells us the effective number of independent latent signals in a dataset!

Video Resource: IITM BS Lecture: Rank and Dimension using Gaussian Elimination 

Official lecture computing matrix rank by row reducing matrices to REF.

4.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Basis Verification:** Check whether $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$ forms a valid basis for $\mathbb{R}^2$. If so, express $\mathbf{v} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ as a linear combination of $\mathcal{B}$.
2. **Finding a Basis for a Subspace:** Find a basis and the dimension for the subspace of $\mathbb{R}^3$:

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \mid x + 2y - z = 0 \right\}$$

3. **Matrix Rank Calculation:** Compute the rank of the 3×4 matrix A by reducing it to REF:

$$A = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 3 & 11 \\ 3 & 6 & 4 & 15 \end{bmatrix}$$

4. **Full Rank vs Rank Deficient:** A dataset matrix $X \in \mathbb{R}^{100 \times 5}$ has 100 samples and 5 features. What is the maximum possible rank of X ? If $\text{rank}(X) = 3$, what does this reveal about the features?
5. **Data Science Application (Collinear Feature Removal):** Given 3 feature vectors:

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{f}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \quad \mathbf{f}_3 = \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix}$$

Find the rank of the matrix $F = [\mathbf{f}_1 \ \mathbf{f}_2 \ \mathbf{f}_3]$, extract a minimal basis of independent feature vectors, and identify which feature is redundant.

Chapter 5

Null Space, Nullity & Linear Mappings

Matrices are not merely static arrays of numbers—they act as dynamic geometric transformations between vector spaces. In this chapter, we explore the **Null Space** of a matrix and formalize **Linear Mappings** (Linear Transformations).

5.1 Null Space and Nullity

Definition: Null Space ($\text{Null}(A)$ or $\ker(A)$)

For a matrix $A \in \mathbb{R}^{m \times n}$, the **Null Space** (or Kernel) of A is the set of all input vectors $\mathbf{x} \in \mathbb{R}^n$ that map to the zero vector $\mathbf{0} \in \mathbb{R}^m$:

$$\text{Null}(A) = \left\{ \mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0} \right\}$$

The dimension of the null space is called the **Nullity** of A :

$$\text{nullity}(A) = \dim(\text{Null}(A))$$

Theorem / Rule: Null Space is a Subspace

The null space $\text{Null}(A)$ is always a valid vector subspace of the domain $\mathbb{R}^n$. To find a basis for $\text{Null}(A)$, reduce $[A \mid \mathbf{0}]$ to REF/RREF, identify free variables, and solve for the basic variables in terms of the free variables.

Video Resource: IITM BS Lecture: The Null Space of a Matrix (Parts 1 & 2) 

Official lectures defining null space, solving homogenous systems $A\mathbf{x} = \mathbf{0}$, and extracting a basis for $\text{Null}(A)$.

5.2 Linear Mappings and Linear Transformations

A function $T : V \rightarrow W$ between two vector spaces V and W is a **Linear Mapping** (or Linear Transformation) if it preserves vector addition and scalar multiplication.

Definition: Linear Transformation Axioms

A mapping $T : V \rightarrow W$ is **Linear** if for all $\mathbf{u}, \mathbf{v} \in V$ and all scalars $c \in \mathbb{R}$:

1. **Additivity:** $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.
2. **Homogeneity (Scalar Scaling):** $T(c\mathbf{v}) = cT(\mathbf{v})$.

Together, this implies $T(c_1\mathbf{u} + c_2\mathbf{v}) = c_1T(\mathbf{u}) + c_2T(\mathbf{v})$. Crucial consequence: $T(\mathbf{0}) = \mathbf{0}$ for any linear transformation.

Data Science Relevance: Linear layers in Deep Learning ($\mathbf{y} = W\mathbf{x}$), dimensionality reduction mappings, and continuous linear projections in SVMs are all formal linear transformations!

Video Resource: IITM BS Lecture: What is a Linear Mapping? (Parts 1 & 2) 

Official lectures defining linear mappings, testing additivity and scalar homogeneity.

Video Resource: IITM BS Lecture: What is a Linear Transformation? 

Official lecture demonstrating matrix-induced transformations $T(\mathbf{x}) = A\mathbf{x}$ and geometric transformations (rotations, reflections, shears).

5.3 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Null Space Calculation:** Find a basis for the null space $\text{Null}(A)$ and determine the nullity $\text{nullity}(A)$ for:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 5 \end{bmatrix}$$

2. **Testing Linearity of Mappings:** Determine whether each function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation:

$$(a) \quad T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} 2x - y \\ x + 3y \end{bmatrix}$$

$$(b) \quad T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x^2 \\ y \end{bmatrix}$$

3. **Matrix-Induced Transformation:** Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $T(\mathbf{x}) = A\mathbf{x}$, where:

$$A = \begin{bmatrix} 1 & 0 \\ -2 & 3 \\ 5 & 4 \end{bmatrix}$$

Compute $T \left(\begin{bmatrix} 3 \\ -1 \end{bmatrix} \right)$.

4. **Homogenous System Solutions:** A matrix $A \in \mathbb{R}^{3 \times 5}$ has $\text{rank}(A) = 3$. What is the dimension of the domain $\mathbb{R}^n$? What is the nullity $\text{nullity}(A)$? Are there non-trivial solutions to $A\mathbf{x} = \mathbf{0}$?
5. **Data Science Application (Lossless Linear Compression Null Space):** A linear projection layer $W \in \mathbb{R}^{10 \times 50}$ compresses a 50-dimensional input vector $\mathbf{x} \in \mathbb{R}^{50}$ into a 10-dimensional representation $\mathbf{y} = W\mathbf{x}$. What is the minimum possible dimension of information lost in the null space $\text{Null}(W)$?

Chapter 6

Matrix Representation, Kernel, Image & Rank-Nullity

Every linear transformation between finite-dimensional vector spaces can be uniquely represented as a matrix multiplication with respect to chosen ordered bases. Furthermore, the fundamental relationship between domain dimension, image dimension, and kernel dimension is governed by the celebrated **Rank-Nullity Theorem**.

6.1 Matrix Representation of Linear Transformations

Let $T : V \rightarrow W$ be a linear transformation, where $\mathcal{B}_V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an ordered basis for domain V , and $\mathcal{B}_W = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ is an ordered basis for codomain W .

Definition: Matrix of Transformation $[T]_{\mathcal{B}_V, \mathcal{B}_W}$

The $m \times n$ matrix representing T with respect to bases $\mathcal{B}_V$ and $\mathcal{B}_W$ is constructed by taking the coordinate vectors of $T(\mathbf{v}_j)$ relative to $\mathcal{B}_W$ as its j -th column:

$$A = [T]_{\mathcal{B}_V, \mathcal{B}_W} = \begin{bmatrix} [T(\mathbf{v}_1)]_{\mathcal{B}_W} & [T(\mathbf{v}_2)]_{\mathcal{B}_W} & \dots & [T(\mathbf{v}_n)]_{\mathcal{B}_W} \end{bmatrix} \in \mathbb{R}^{m \times n}$$

Then for any $\mathbf{x} \in V$, coordinate transformation is matrix-vector multiplication: $[T(\mathbf{x})]_{\mathcal{B}_W} = A[\mathbf{x}]_{\mathcal{B}_V}$.

Video Resource: IITM BS Lecture: Linear Transformations, Ordered Bases & Matrices



Official lecture constructing transformation matrices relative to non-standard ordered bases.

6.2 Kernel and Image of Linear Transformations

Definition: Kernel ($\ker(T)$) and Image ($\text{im}(T)$)

- **Kernel (Null Space):** The set of domain vectors mapping to zero:

$$\ker(T) = \left\{ \mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}_W \right\} \subseteq V$$

- **Image (Range / Column Space):** The set of all output vectors in codomain W :

$$\text{im}(T) = \left\{ \mathbf{w} \in W \mid \exists \mathbf{v} \in V \text{ such that } T(\mathbf{v}) = \mathbf{w} \right\} \subseteq W$$

Video Resource: IITM BS Lecture: Image & Kernel of Linear Transformations



Official lecture defining image and kernel subspaces and their geometric interpretations.

Video Resource: IITM BS Lecture: Finding Bases for Kernel and Image



Official lecture computing bases for $\ker(T)$ and $\text{im}(T)$ using REF pivot columns.

6.3 The Rank-Nullity Theorem

Theorem / Rule: The Fundamental Rank-Nullity Theorem

For any linear transformation $T : V \rightarrow W$ on a finite-dimensional domain V :

$$\dim(\ker(T)) + \dim(\text{im}(T)) = \dim(V)$$

In matrix terms for $A \in \mathbb{R}^{m \times n}$:

$$\text{nullity}(A) + \text{rank}(A) = n \quad (\text{number of columns / input features})$$

Data Science Relevance: The Rank-Nullity Theorem guarantees conservation of dimension when passing data through linear autoencoders and neural network linear layers.

6.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Matrix Representation Construction:** Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} 3x + y \\ 2x - 4y \end{bmatrix}$. Find the matrix representation $A = [T]_{\mathcal{E}}$ relative to the standard basis $\mathcal{E}$.
2. **Non-Standard Basis Matrix Representation:** Find the matrix $[T]_{\mathcal{B}}$ for the linear operator $T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x + y \\ 2x - y \end{bmatrix}$ using the ordered basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$.
3. **Bases for Kernel and Image:** Given linear transformation matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$:
 - (a) Find a basis for $\text{im}(A)$ (Column Space).
 - (b) Find a basis for $\text{ker}(A)$ (Null Space).
4. **Rank-Nullity Verification:** Verify the Rank-Nullity Theorem for the matrix A in Exercise 3 by showing $\text{nullity}(A) + \text{rank}(A) = 3$.
5. **Data Science Application (Linear Autoencoder Information Loss):** A linear autoencoder compresses 100-dimensional audio feature vectors $\mathbf{x} \in \mathbb{R}^{100}$ into a bottleneck feature space of dimension 20 using matrix $W \in \mathbb{R}^{20 \times 100}$. If $\text{rank}(W) = 18$, calculate:
 - (a) The dimension of the compressed code space ($\dim(\text{im}(W))$).
 - (b) The dimension of unrecoverable lost features ($\dim(\text{ker}(W))$).

Chapter 7

Matrix Similarity, Affine Spaces, Norms & Inner Products

Geometric intuition in high-dimensional space requires measuring lengths, distances, and angles between vectors. In this chapter, we explore **Matrix Similarity** (change of basis), **Affine Spaces**, and formalize geometric measurement using **Norms** and **Inner Products**.

7.1 Matrix Similarity and Change of Basis

Two square matrices $A, B \in \mathbb{R}^{n \times n}$ are **Similar** if they represent the exact same linear operator $T : V \rightarrow V$ under two different choices of basis.

Theorem / Rule: Matrix Similarity

Matrix B is **similar** to A (written $B \sim A$) if there exists an invertible change-of-basis matrix $P \in \mathbb{R}^{n \times n}$ such that:

$$B = P^{-1}AP$$

Invariants under Similarity: Similar matrices share the exact same:

1. Determinant: $\det(B) = \det(A)$.
2. Trace: $\text{tr}(B) = \text{tr}(A)$.
3. Rank: $\text{rank}(B) = \text{rank}(A)$.
4. Eigenvalues and characteristic polynomial.

Core Intuition: Similar Matrices = Same Transformation, Different Perspectives

If matrix B is similar to matrix A , they are fundamentally doing the exact same thing to space!

- Imagine you are giving a robot directions. You might say “move 2 meters North, 1 meter East” (Standard Basis).
- Your friend, standing at an angle, might say “move 1 meter Forward, 2 meters Right” (New Basis).
- You are both describing the **exact same physical movement**, just using different coordinate systems.
- The matrix P translates between your languages. $B = P^{-1}AP$ means: translate to your friend’s language (P), do the movement (A), and translate back (P^{-1}).

Video Resource: IITM BS Lecture: Equivalence & Similarity of Matrices 

Official lecture explaining change-of-basis transformations and similarity invariants.

7.2 Affine Subspaces and Mappings

An **Affine Subspace** (or Flat) is a translated vector subspace: $\mathcal{A} = \mathbf{x}_0 + W$, where $\mathbf{x}_0 \in V$ is a fixed translation point and $W \subseteq V$ is a vector subspace.

Definition: Affine Subspace vs Vector Subspace

Unlike vector subspaces, an affine subspace $\mathbf{x}_0 + W$ does **NOT** need to pass through the origin $\mathbf{0}$ (unless $\mathbf{x}_0 \in W$). For example, lines and planes in $\mathbb{R}^3$ that do not pass through $(0, 0, 0)$ are affine subspaces.

Video Resource: IITM BS Lecture: Affine Subspaces & Mappings 

Official lecture contrasting vector subspaces with shifted affine spaces and hyperplanes.

7.3 Inner Products and Norms

An **Inner Product** $\langle \mathbf{u}, \mathbf{v} \rangle$ generalizes the dot product, providing a rigorous concept of length and angle for any vector space.

Definition: Inner Product Space Axioms

An **Inner Product** $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfies:

1. **Symmetry:** $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$.
2. **Linearity in first argument:** $\langle c\mathbf{u} + d\mathbf{v}, \mathbf{w} \rangle = c\langle \mathbf{u}, \mathbf{w} \rangle + d\langle \mathbf{v}, \mathbf{w} \rangle$.
3. **Positive Definiteness:** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, and $\langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = \mathbf{0}$.

Every inner product defines an **Induced Norm**: $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

Key Formula / Algorithm: Cauchy-Schwarz Inequality & Angle

For any inner product space:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

The angle θ between non-zero vectors $\mathbf{u}, \mathbf{v}$ is defined by:

$$\cos(\theta) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

Data Science Relevance: L_1 norm (Manhattan distance, Lasso regularization), L_2 norm (Euclidean distance, Ridge regularization), and Cosine Similarity are core tools in clustering (K-Means), text retrieval, and model loss formulation!

Video Resource: IITM BS Lecture: Lengths, Angles & Inner Products



Official lecture deriving lengths, unit vectors $\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$, and Cauchy-Schwarz inequality.

Video Resource: IITM BS Lecture: Inner Products and Norms on a Vector Space



Official lecture introducing weighted inner products $\langle \mathbf{u}, \mathbf{v} \rangle_Q = \mathbf{u}^T Q \mathbf{v}$ with positive-definite Q .

7.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Matrix Similarity Check:** Given $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}$:
 - (a) Compute $\det(A)$ and $\det(B)$.
 - (b) Compute $\text{tr}(A)$ and $\text{tr}(B)$.
 - (c) Are A and B similar matrices?
2. **Weighted Inner Product:** Define a weighted inner product on $\mathbb{R}^2$ by $\langle \mathbf{u}, \mathbf{v} \rangle_Q = 2u_1v_1 + 5u_2v_2$.
 - (a) Compute $\left\langle \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \end{bmatrix} \right\rangle_Q$.
 - (b) Compute the induced norm $\left\| \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\|_Q$.

3. **Cosine Similarity:** In natural language processing, word embeddings for two documents are:

$$\mathbf{d}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{d}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Compute the Cosine Similarity $\cos(\theta) = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{\|\mathbf{d}_1\| \|\mathbf{d}_2\|}$.

4. **Cauchy-Schwarz Verification:** For $\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in standard $\mathbb{R}^2$, verify that $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$.
5. **Data Science Application (L_1 vs L_2 Norm Regularization):** A weights vector in a sparse ML model is $\mathbf{w} = \begin{bmatrix} 3 \\ -4 \\ 0 \\ 12 \end{bmatrix}$.
 - (a) Calculate the L_1 norm $\|\mathbf{w}\|_1 = \sum |w_i|$ (Lasso penalty).
 - (b) Calculate the L_2 norm $\|\mathbf{w}\|_2 = \sqrt{\sum w_i^2}$ (Ridge penalty).

Chapter 8

Orthogonality, Gram-Schmidt Process & Projections

Orthogonality is the mathematical formalization of perpendicularity. Orthogonal vectors provide independent, uncorrelated coordinates. In this chapter, we master **Orthonormal Bases**, **Orthogonal Projections**, and the **Gram-Schmidt Process**.

8.1 Orthogonality and Orthonormal Bases

Definition: Orthogonality & Orthonormal Basis

- Two vectors $\mathbf{u}, \mathbf{v}$ are **Orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ (written $\mathbf{u} \perp \mathbf{v}$).
- A set of vectors $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ is an **Orthonormal Basis (ONB)** if:

$$\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \quad (\text{Unit Norm: } \|\mathbf{q}_i\| = 1) \\ 0 & \text{if } i \neq j \quad (\text{Mutually Orthogonal}) \end{cases}$$

Theorem / Rule: Coordinate Computation with ONB

If $\mathcal{B} = \{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ is an ONB for V , the coordinates of any $\mathbf{v} \in V$ are obtained simply by taking inner products without solving linear systems!

$$\mathbf{v} = \sum_{i=1}^n \langle \mathbf{v}, \mathbf{q}_i \rangle \mathbf{q}_i = \langle \mathbf{v}, \mathbf{q}_1 \rangle \mathbf{q}_1 + \dots + \langle \mathbf{v}, \mathbf{q}_n \rangle \mathbf{q}_n$$

Video Resource: IITM BS Lecture: Orthogonality & Linear Independence 

Official lecture proving that any set of non-zero mutually orthogonal vectors is automatically linearly independent.

Video Resource: IITM BS Lecture: What is an Orthonormal Basis? 

Official lecture defining ONB and showing simple coordinate extraction via inner products.

8.2 Orthogonal Projections

The **Orthogonal Projection** of a vector $\mathbf{v}$ onto a subspace W finds the unique point $\mathbf{p} = \text{proj}_W(\mathbf{v}) \in W$ that minimizes the distance $\|\mathbf{v} - \mathbf{p}\|$.

Key Formula / Algorithm: Projection onto Subspace W with ONB $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$

$$\text{proj}_W(\mathbf{v}) = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{q}_i \rangle \mathbf{q}_i$$

The error vector $\mathbf{e} = \mathbf{v} - \text{proj}_W(\mathbf{v})$ is strictly orthogonal to W ($\mathbf{e} \perp W$).

Video Resource: IITM BS Lecture: Projections Using Inner Products 

Official lecture deriving projection formulas, best approximation theorem, and error orthogonality.

8.3 The Gram-Schmidt Orthogonalization Process

The **Gram-Schmidt Process** converts an arbitrary linearly independent basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ into an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k\}$.

Key Formula / Algorithm: Gram-Schmidt Algorithm

1. **First Vector:** $\mathbf{u}_1 = \mathbf{v}_1$, $\mathbf{q}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$ 2. **Second Vector:** Subtract projection onto $\mathbf{q}_1$:

$$\mathbf{u}_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{q}_1 \rangle \mathbf{q}_1, \quad \mathbf{q}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|}$$

3. **k -th Vector:** Subtract projections onto all previous $\mathbf{q}_j$:

$$\mathbf{u}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \langle \mathbf{v}_k, \mathbf{q}_j \rangle \mathbf{q}_j, \quad \mathbf{q}_k = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|}$$

Data Science Relevance: The Gram-Schmidt process is the foundation of **QR Factorization** ($A = QR$) used in stable Ordinary Least Squares regression and numerical eigenvalue algorithms!

Video Resource: IITM BS Lecture: The Gram-Schmidt Process 

Official lecture executing Gram-Schmidt step-by-step to construct orthonormal bases.

Video Resource: IITM BS Lecture: Orthogonal Transformations & Rotations 

Official lecture introducing orthogonal matrices $Q^T Q = I$ (preserves lengths and angles) and rotation matrices.

8.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Projection onto a Vector:** Compute the orthogonal projection of vector $\mathbf{v} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ onto the line spanned by $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.
2. **Gram-Schmidt Process (2×2):** Use the Gram-Schmidt process to convert the basis $\mathcal{B} = \left\{ \mathbf{v}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ into an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2\}$.
3. **Gram-Schmidt Process (3×3):** Apply Gram-Schmidt to orthogonalize $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.
4. **Orthogonal Matrix Properties:** Let $Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ be a 2D rotation matrix.
 - (a) Compute $Q^T Q$.
 - (b) Show that $\|Q\mathbf{x}\| = \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^2$ (preserves norm).
5. **Data Science Application (Least Squares via Projection):** In linear regression, the best fit prediction vector $\hat{\mathbf{y}} = X\mathbf{w}$ is the orthogonal projection of target vector $\mathbf{y}$ onto the column space of feature matrix X :

$$\hat{\mathbf{y}} = P\mathbf{y}, \quad \text{where } P = X(X^T X)^{-1} X^T$$

Show that the projection matrix P is symmetric ($P^T = P$) and idempotent ($P^2 = P$).

Chapter 9

Multivariable Functions, Partial Derivatives & Gradient

Transitioning from 1D single-variable calculus to **Multivariable Calculus** allows us to model functions with multiple input features: $f(\mathbf{x}) = f(x_1, x_2, \dots, x_n)$. In this chapter, we master **Partial Derivatives**, **Directional Derivatives**, and the **Gradient Vector** ∇f .

9.1 Multivariable Functions and Visualizations

A scalar-valued multivariable function maps a feature vector in $\mathbb{R}^n$ to a real number output:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

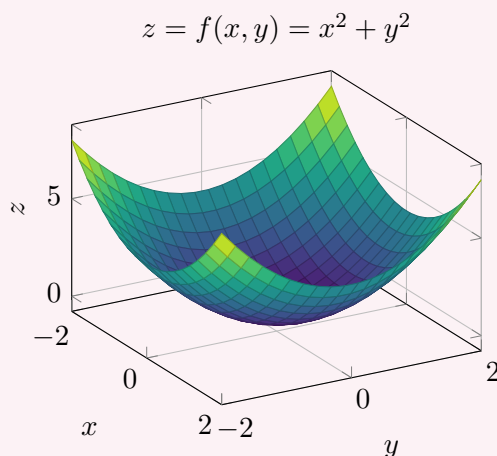
Definition: Level Sets and Contour Curves

For $f(x, y)$, a **Level Curve** (or Contour) is the set of all points (x, y) where the function takes a constant value c :

$$\{(x, y) \in \mathbb{R}^2 \mid f(x, y) = c\}$$

Core Intuition: 3D Surface Loss Bowl and Contour Map

Below is a 3D visualization of a convex loss surface $z = f(x, y) = x^2 + y^2$ alongside its 2D concentric circular contour map.



Video Resource: IITM BS Lecture: Multivariable Functions & Visualization 

Official lecture visualizing 3D surface plots $z = f(x, y)$ and 2D contour lines.

9.2 Partial Derivatives

The **Partial Derivative** of $f(x_1, x_2, \dots, x_n)$ with respect to x_i measures the rate of change of f along the x_i coordinate axis, holding all other variables constant.

Theorem / Rule: Partial Derivative Calculation

To compute $\frac{\partial f}{\partial x_i}$ (or f_{x_i}), treat all variables except x_i as constants and differentiate using standard 1D calculus rules.

$$\frac{\partial f}{\partial x_1} = \lim_{h \rightarrow 0} \frac{f(x_1 + h, x_2, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h}$$

Example: Computing Partial Derivatives

For $f(x, y) = x^3y^2 + 5x^2 - 4y$:

$$\frac{\partial f}{\partial x} = 3x^2y^2 + 10x$$

$$\frac{\partial f}{\partial y} = 2x^3y - 4$$

Video Resource: IITM BS Lecture: Partial Derivatives 

Official lecture defining partial derivatives geometrically as tangent slopes along coordinate grid slices.

9.3 The Gradient Vector and Directional Derivatives

The **Gradient** $\nabla f(\mathbf{x})$ packs all first-order partial derivatives into a single column vector.

Definition: The Gradient Vector (∇f)

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **Gradient Vector** is:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n$$

Key Formula / Algorithm: Directional Derivative via Gradient

The **Directional Derivative** $D_{\mathbf{u}}f(\mathbf{x})$ measures the rate of change of f in the direction of a **unit vector** $\mathbf{u}$ ($\|\mathbf{u}\| = 1$):

$$D_{\mathbf{u}}f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{u} = \|\nabla f(\mathbf{x})\| \cos \theta$$

Data Science Relevance: The gradient $\nabla f(\mathbf{x})$ points in the direction of **steepest increase** of the function, while $-\nabla f(\mathbf{x})$ points in the direction of **steepest descent** the core mechanism behind updating neural network weights during Gradient Descent training!

Video Resource: IITM BS Lecture: Directional Derivatives



Official lecture defining rate of change along arbitrary unit vector directions.

Video Resource: IITM BS Lecture: Directional Derivatives in Terms of Gradient



Official lecture proving $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ and geometric properties of gradients.

9.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **First-Order Partial Derivatives:** Compute $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for:

$$f(x, y) = x^2 y^3 + e^{2x} \sin(y)$$

2. **Gradient Vector Evaluation:** Compute the gradient vector $\nabla f(1, 2)$ for:

$$f(x, y) = 4x^2 y - 3y^2 + 5x$$

3. **Directional Derivative:** Calculate the directional derivative $D_{\mathbf{u}}f(2, 1)$ for $f(x, y) = x^2 + 3xy + y^2$ in the direction of vector $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. (Remember to normalize $\mathbf{v}$ into a unit vector $\mathbf{u}$).
4. **Steepest Ascent Direction:** At the point $(1, -1)$, in which unit vector direction does $f(x, y) = x^3 y - 2xy^2$ increase most rapidly? What is the maximum rate of increase?
5. **Data Science Application (MSE Loss Gradient):** In linear regression with 1 sample (x_i, y_i) , the Mean Squared Error loss as a function of slope m and bias b is:

$$L(m, b) = (mx_i + b - y_i)^2$$

Compute the gradient vector $\nabla L(m, b) = \begin{bmatrix} \frac{\partial L}{\partial m} \\ \frac{\partial L}{\partial b} \end{bmatrix}$.

Chapter 10

Tangent Planes, Steepest Descent & Critical Points

Optimization is the heart of Machine Learning. Training a model means finding parameter values that minimize a loss function $L(\mathbf{w})$. In this chapter, we master **Tangent Planes/Hyperplanes**, the **Gradient Descent Update Rule**, and **Critical Points** ($\nabla f(\mathbf{x}) = 0$).

10.1 Tangent Planes and Hyperplanes

The 1D tangent line generalizes in higher dimensions to a **Tangent Plane** (for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$) or a **Tangent Hyperplane** (for $f : \mathbb{R}^n \rightarrow \mathbb{R}$).

Definition: Tangent Plane Formula

For a surface $z = f(x, y)$, the tangent plane at point (x_0, y_0, z_0) where $z_0 = f(x_0, y_0)$ is given by:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

In general for $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the linear approximation (tangent hyperplane) at $\mathbf{x}_0$ is:

$$L(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0)$$

Video Resource: IITM BS Lecture: Tangents for Scalar-Valued Functions 

Official lecture deriving tangent plane equations and linear approximations.

Video Resource: IITM BS Lecture: Finding the Tangent Hyperplane 

Official lecture extending tangent planes to n -dimensional hyperplanes $\nabla f(\mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0) = 0$.

10.2 Steepest Descent and Gradient Descent Algorithm

Theorem / Rule: Direction of Steepest Descent

- **Steepest Ascent:** The unit vector direction along which $f(\mathbf{x})$ increases fastest is $\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$.
- **Steepest Descent:** The unit vector direction along which $f(\mathbf{x})$ decreases fastest is $-\frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|}$.

Key Formula / Algorithm: Gradient Descent Update Rule

To minimize a cost function $L(\mathbf{w})$, start at an initial point $\mathbf{w}^{(0)}$ and iteratively update:

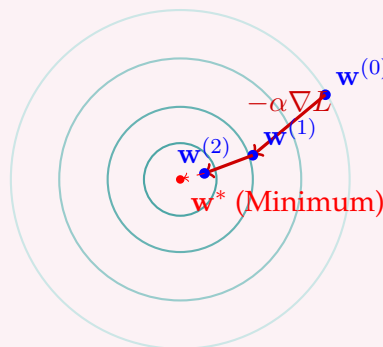
$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} - \alpha \nabla L(\mathbf{w}^{(k)})$$

where $\alpha > 0$ is the **Learning Rate** (step size).

Data Science Relevance: Gradient Descent (and its variants like Adam, RMSprop, SGD) is used to train almost all modern Machine Learning models, from Logistic Regression to 70-Billion parameter Large Language Models!

Core Intuition: Gradient Descent Step Trajectory on Contour Map

Gradient descent takes steps perpendicular to contour curves, moving in the direction of $-\nabla f$ toward the global minimum.



Video Resource: IITM BS Lecture: The Direction of Steepest Ascent / Descent



Official lecture proving why $-\nabla f$ is the optimal search direction for loss minimization.

10.3 Critical Points

Definition: Critical Points

An interior point $\mathbf{x}_0$ in the domain of f is a **Critical Point** (or Stationary Point) if:

$$\nabla f(\mathbf{x}_0) = \mathbf{0} \iff \frac{\partial f}{\partial x_1}(\mathbf{x}_0) = 0, \frac{\partial f}{\partial x_2}(\mathbf{x}_0) = 0, \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}_0) = 0$$

Critical points can be **Local Minima**, **Local Maxima**, or **Saddle Points**.

Video Resource: IITM BS Lecture: Critical Points for Multivariable Functions 

Official lecture finding critical points by setting all first partial derivatives to zero simultaneously.

10.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Tangent Plane Equation:** Find the equation of the tangent plane to the surface $z = x^2 + 3y^2 - 4x$ at the point $(x_0, y_0) = (2, 1)$.
2. **Gradient Descent Iteration Step:** A 2D loss function is $L(w_1, w_2) = w_1^2 + 4w_2^2$.
 - (a) Compute the gradient $\nabla L(w_1, w_2)$.
 - (b) Starting from $\mathbf{w}^{(0)} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ with learning rate $\alpha = 0.1$, compute the next weight vector $\mathbf{w}^{(1)}$.

3. **Finding Critical Points (2×2):** Find all critical points of the function:

$$f(x, y) = x^3 + y^3 - 3xy$$

by setting $f_x = 0$ and $f_y = 0$.

4. **Linear Approximation:** Use the tangent plane linear approximation of $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$ to estimate $f(3.06, 3.92)$.
5. **Data Science Application (Least Squares Gradient Descent Update):** For a simple regression model with loss $L(w) = (2w - 6)^2$, start at $w^{(0)} = 0$ with learning rate $\alpha = 0.05$. Perform 2 steps of Gradient Descent to find $w^{(1)}$ and $w^{(2)}$. Show that w moves towards the true minimum $w^* = 3$.

Chapter 11

Hessian Matrix, Multivariable Extrema & Differentiability

To determine whether a critical point $\mathbf{x}_0$ is a local minimum, local maximum, or saddle point, we analyze second-order partial derivatives. In this chapter, we master the **Hessian Matrix**, the **Second Derivative Test**, and formal **Multivariable Differentiability**.

11.1 Second-Order Partial Derivatives and Clairaut's Theorem

For $f(x, y)$, there are four second-order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial^2 f}{\partial y \partial x}, \quad f_{yx} = \frac{\partial^2 f}{\partial x \partial y}$$

Theorem / Rule: Clairaut's Theorem (Symmetry of Mixed Partial)

If f_{xy} and f_{yx} are continuous on an open set containing $\mathbf{x}_0$, then the mixed partials are equal:

$$f_{xy} = f_{yx}$$

11.2 The Hessian Matrix

Definition: The Hessian Matrix $H(f)$

The **Hessian Matrix** $H(f)(\mathbf{x})$ is the symmetric matrix of second-order partial derivatives:

$$H(f)(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \in \mathbb{R}^{n \times n}$$

Video Resource: IITM BS Lecture: Higher Order Partial Derivatives & Hessian Matrix



Official lecture constructing Hessian matrices for 2D and 3D functions.

11.3 Second Derivative Test for Local Extrema

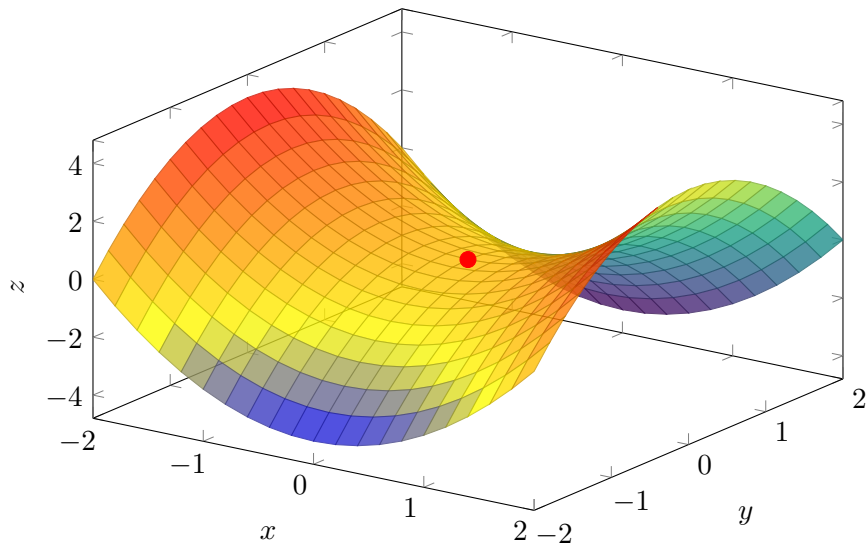
Theorem / Rule: Second Derivative Test for $f(x, y)$

Let (x_0, y_0) be a critical point ($\nabla f(x_0, y_0) = \mathbf{0}$). Define the Hessian determinant D :

$$D = \det(H(f)(x_0, y_0)) = f_{xx}f_{yy} - (f_{xy})^2$$

1. If $D > 0$ and $f_{xx} > 0$ (Hessian is Positive Definite): (x_0, y_0) is a **Local Minimum**.
2. If $D > 0$ and $f_{xx} < 0$ (Hessian is Negative Definite): (x_0, y_0) is a **Local Maximum**.
3. If $D < 0$ (Hessian is Indefinite): (x_0, y_0) is a **Saddle Point**.
4. If $D = 0$: Test is **Inconclusive**.

Visualization of a Saddle Point: $f(x, y) = x^2 - y^2$



Data Science Relevance: In Optimization and Deep Learning, Newton-Raphson optimization uses the inverse Hessian $H^{-1}\nabla f$ for second-order gradient updates. Convex loss functions (where $H(L)$ is positive semi-definite everywhere) guarantee that any local minimum is a global minimum!

Video Resource: IITM BS Lecture: Hessian Matrix & Local Extrema for $f(x, y)$



Official lecture classifying 2D critical points using the Hessian discriminant D .

Video Resource: IITM BS Lecture: Hessian Matrix for 3D Functions $f(x, y, z)$



Official lecture extending local extrema classification to 3D via leading principal minors of $H(f)$.

11.4 Multivariable Differentiability

Unlike 1D functions, the existence of partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ does **NOT** guarantee that a multivariable function is continuous or differentiable!

Definition: Formal Multivariable Differentiability

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **Differentiable** at $\mathbf{x}_0$ if there exists a linear mapping (the derivative total transformation J) such that:

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - \nabla f(\mathbf{x}_0)^T \mathbf{h}|}{\|\mathbf{h}\|} = 0$$

Sufficient Condition: If all first partial derivatives $\frac{\partial f}{\partial x_i}$ exist and are **continuous** near $\mathbf{x}_0$, then f is guaranteed to be differentiable at $\mathbf{x}_0$.

Video Resource: IITM BS Lecture: Differentiability for Multivariable Functions 

Official lecture contrasting partial derivative existence with formal multivariable differentiability.

11.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Hessian Matrix Computation:** Compute the Hessian matrix $H(f)(x, y)$ for $f(x, y) = x^3 - 3xy^2 + y^4$. Evaluate it at the point $(1, 2)$.
2. **Classifying 2D Critical Points:** Find all critical points of $f(x, y) = x^3 + y^3 - 3xy$ (from Week 10 exercise: $(0, 0)$ and $(1, 1)$), and classify each point as a local minimum, local maximum, or saddle point using the Hessian test.
3. **Saddle Point Identification:** Show that the origin $(0, 0)$ is a saddle point for the hyperbolic paraboloid $f(x, y) = x^2 - y^2$ by analyzing the eigenvalues or determinant of $H(f)(0, 0)$.
4. **Classifying 3D Critical Points:** For $f(x, y, z) = x^2 + 2y^2 + 3z^2 - 4x + 8y - 12z$, find the critical point and construct the 3×3 Hessian matrix to classify it.
5. **Data Science Application (Convexity of Mean Squared Error):** The OLS Linear Regression loss function for 1 feature w is $L(w) = \frac{1}{N} \sum_{i=1}^N (wx_i - y_i)^2$.
 - (a) Compute the second derivative $\frac{d^2L}{dw^2}$.
 - (b) Show that $\frac{d^2L}{dw^2} \geq 0$ for all w , proving that the loss surface is strictly convex and has a unique global minimum.

Chapter 12

Eigenvalues, Eigenvectors, Diagonalization & SVD

Eigenvalues and Eigenvectors form the ultimate cornerstone of modern linear algebra and data science. In this capstone chapter, we master **Eigen-decomposition**, **Matrix Diagonalization**, and **Singular Value Decomposition (SVD)**.

12.1 Eigenvalues and Eigenvectors

When a matrix $A \in \mathbb{R}^{n \times n}$ acts on a vector $\mathbf{v}$, it typically changes both the magnitude and direction of $\mathbf{v}$. However, certain non-zero vectors maintain their original direction and are merely scaled by a factor λ .

Definition: Eigenvalue Problem

A non-zero vector $\mathbf{v} \neq \mathbf{0}$ is an **Eigenvector** of A with corresponding **Eigenvalue** $\lambda \in \mathbb{R}$ (or $\mathbb{C}$) if:

$$A\mathbf{v} = \lambda\mathbf{v}$$

Equivalently:

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

Theorem / Rule: The Characteristic Equation

Since $\mathbf{v} \neq \mathbf{0}$, the matrix $(A - \lambda I)$ must be singular ($\det(A - \lambda I) = 0$). 1. **Characteristic Polynomial:** $p(\lambda) = \det(A - \lambda I) = 0$. Solve for roots $\lambda_1, \lambda_2, \dots, \lambda_n$. 2. **Eigenspace:** For each λ_i , solve $(A - \lambda_i I)\mathbf{v} = \mathbf{0}$ to find the basis of eigenvectors $\text{Null}(A - \lambda_i I)$.

Video Resource: 3Blue1Brown: Essence of Linear Algebra (Eigenvalues & Eigenvectors)



Visual intuition showing eigenvectors as axes of rotation/scaling during linear transformations.

12.2 Matrix Diagonalization

Definition: Diagonalization Theorem

An $n \times n$ matrix A is **Diagonalizable** if it has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$. Form eigenvector matrix $P = [\mathbf{v}_1 \dots \mathbf{v}_n]$ and diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then:

$$A = PDP^{-1} \iff D = P^{-1}AP$$

Matrix Powers: $A^k = PD^kP^{-1}$, making high-power matrix computations $\mathcal{O}(n)$ instead of $\mathcal{O}(n^k)$!

Theorem / Rule: Spectral Theorem for Symmetric Matrices

If $A = A^T$ is a real symmetric matrix, then: 1. All eigenvalues of A are real numbers. 2. Eigenvectors corresponding to distinct eigenvalues are **mutually orthogonal**. 3. A is **orthogonally diagonalizable**: $A = QDQ^T$, where Q is an orthogonal matrix ($Q^TQ = I$).

12.3 Singular Value Decomposition (SVD)

While Eigen-decomposition requires square matrices $A \in \mathbb{R}^{n \times n}$, **Singular Value Decomposition (SVD)** applies to ANY rectangular matrix $A \in \mathbb{R}^{m \times n}$.

Key Formula / Algorithm: Singular Value Decomposition ($A = U\Sigma V^T$)

Every matrix $A \in \mathbb{R}^{m \times n}$ can be factored into:

$$A = U\Sigma V^T$$

where:

- $U \in \mathbb{R}^{m \times m}$: Left singular vectors (orthonormal eigenvectors of AA^T).
- $\Sigma \in \mathbb{R}^{m \times n}$: Diagonal matrix of singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ ($\sigma_i = \sqrt{\lambda_i(A^T A)}$).
- $V \in \mathbb{R}^{n \times n}$: Right singular vectors (orthonormal eigenvectors of $A^T A$).

```

1 import numpy as np
2
3 # 1. Eigenvalue Decomposition
4 A = np.array([[4.0, 1.0],
5               [2.0, 3.0]])
6
7 eigenvalues, eigenvectors = np.linalg.eig(A)
8 print("Eigenvalues:", eigenvalues)
9 print("Eigenvectors:\n", eigenvectors)
10
11 # Verify A @ v = lambda * v for the first eigenvector
12 v0 = eigenvectors[:, 0]
13 lam0 = eigenvalues[0]
14 print("A @ v0:      ", A @ v0)
15 print("lambda * v0:", lam0 * v0)

```

```
16
17 # 2. Singular Value Decomposition (SVD)
18 X = np.array([[3.0, 1.0],
19               [1.0, 3.0],
20               [0.0, 2.0]])
21
22 U, S, Vt = np.linalg.svd(X)
23 print("\nLeft Singular Vectors U shape:", U.shape)
24 print("Singular Values S:", S)
25 print("Right Singular Vectors V^T shape:", Vt.shape)
```

Listing 12.1: NumPy Implementation of Eigen-decomposition and SVD

Video Resource: StatQuest: Singular Value Decomposition (SVD) & PCA 

Step-by-step visual explanation of SVD matrix factorization and dimensionality reduction.

Video Resource: IITM BS Lecture: Mathematics for Data Science II Review 

Official comprehensive course summary reviewing Linear Algebra and Multivariable Calculus.

12.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Eigenvalues & Eigenvectors (2×2):** Find the eigenvalues λ_1, λ_2 and corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ for the matrix:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

2. **Matrix Diagonalization ($A = PDP^{-1}$):** Using the eigenvalues and eigenvectors from Exercise 1, construct the matrices P and D such that $A = PDP^{-1}$. Compute A^4 using PD^4P^{-1} .
3. **Symmetric Matrix Spectral Decomposition:** Verify that $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ is symmetric. Find its real eigenvalues and construct an orthogonal matrix Q ($Q^T Q = I$) such that $A = QDQ^T$.
4. **Singular Values Calculation:** Given matrix $A = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$, compute the matrix $A^T A$, find its eigenvalues λ_i , and determine the singular values σ_1, σ_2 of A .
5. **Comprehensive Course Review Question (PCA via SVD):** A data matrix $X \in \mathbb{R}^{N \times d}$ has covariance matrix $C = \frac{1}{N} X^T X$.
 - (a) Explain why the principal component directions are the eigenvectors of C .
 - (b) If the eigenvalues of C are $\lambda_1 = 80$, $\lambda_2 = 15$, $\lambda_3 = 5$, what percentage of total dataset variance is explained by keeping the first 2 principal components?

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Solutions to Practice Exercises

This appendix contains complete, step-by-step mathematical solutions for all practice exercises across Weeks 1 to 12.

Week 1: Vectors, Matrices & Determinants

Detailed Solutions

1. Vector Dot Product & Norms

- (a) Dot Product:

$$\mathbf{u} \cdot \mathbf{v} = (2)(1) + (3)(-2) + (-1)(4) = 2 - 6 - 4 = -8$$

- (b) Norms:

$$\|\mathbf{u}\| = \sqrt{2^2 + 3^2 + (-1)^2} = \sqrt{4 + 9 + 1} = \sqrt{14}$$

$$\|\mathbf{v}\| = \sqrt{1^2 + (-2)^2 + 4^2} = \sqrt{1 + 4 + 16} = \sqrt{21}$$

2. Matrix Multiplication

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 1(0) + 2(2) & 1(-1) + 2(5) \\ 3(0) + 4(2) & 3(-1) + 4(5) \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 8 & 17 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 0(1) - 1(3) & 0(2) - 1(4) \\ 2(1) + 5(3) & 2(2) + 5(4) \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ 17 & 24 \end{bmatrix}$$

Since $\begin{bmatrix} 4 & 9 \\ 8 & 17 \end{bmatrix} \neq \begin{bmatrix} -3 & -4 \\ 17 & 24 \end{bmatrix}$, matrix multiplication is non-commutative ($AB \neq BA$).

3. Matrix-Vector System Formulation

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 4 & -1 & 2 \\ -1 & 5 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 7 \\ 3 \\ 10 \end{bmatrix}$$

Augmented Matrix $[A \mid \mathbf{b}]$:

$$[A \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 4 & -1 & 2 & 3 \\ -1 & 5 & 3 & 10 \end{array} \right]$$

4. Determinant Calculation Expand M along row 1:

$$\det(M) = 3 \begin{vmatrix} 4 & 5 \\ 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 5 \\ 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 0 & 4 \\ 1 & 2 \end{vmatrix}$$

$$\det(M) = 3(4 - 10) - 1(0 - 5) + 2(0 - 4) = 3(-6) - 1(-5) + 2(-4) = -18 + 5 - 8 = -21$$

Since $\det(M) = -21 \neq 0$, the matrix M is **invertible**.

5. Data Science Application: Covariance Determinant

$$\det(\Sigma) = \begin{vmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{vmatrix} = \sigma_1^2 \sigma_2^2 - \sigma_{12}^2$$

$$\det(\Sigma) = (4)(9) - (3)^2 = 36 - 9 = \mathbf{27}$$

Week 2: Cramer's Rule, Row Echelon Form & Gaussian Elimination

Detailed Solutions

1. Cramer's Rule (2×2)

$$\det(A) = \begin{vmatrix} 3 & 4 \\ 1 & -2 \end{vmatrix} = 3(-2) - 4(1) = -6 - 4 = \mathbf{-10}$$

$$\det(A_1(\mathbf{b})) = \begin{vmatrix} 10 & 4 \\ -3 & -2 \end{vmatrix} = 10(-2) - 4(-3) = -20 + 12 = -8 \implies x_1 = \frac{-8}{-10} = \mathbf{0.8}$$

$$\det(A_2(\mathbf{b})) = \begin{vmatrix} 3 & 10 \\ 1 & -3 \end{vmatrix} = 3(-3) - 10(1) = -9 - 10 = -19 \implies x_2 = \frac{-19}{-10} = \mathbf{1.9}$$

2. Identifying REF

- **(a) REF:** Pivots are at (1, 1) and (2, 2), entry below pivots is 0, zero row is at bottom.
- **(b) REF:** Upper triangular with pivots at (1, 1), (2, 2), (3, 3).
- **(c) Neither:** The pivot in row 2 is in column 1, which is to the left of the pivot in row 1 (column 2).

3. Gaussian Elimination (3×3) Augmented matrix $[A \mid \mathbf{b}]$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & 3 & 4 & 20 \\ 4 & 3 & 2 & 16 \end{array} \right]$$

1. $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 4R_1$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -1 & 2 & 4 \\ 0 & -5 & -2 & -16 \end{array} \right]$$

2. $R_3 \rightarrow R_3 - 5R_2$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -1 & 2 & 4 \\ 0 & 0 & -12 & -36 \end{array} \right]$$

3. Back-Substitution:

- Row 3: $-12x_3 = -36 \implies \mathbf{x_3 = 3}$.
- Row 2: $-x_2 + 2(3) = 4 \implies -x_2 + 6 = 4 \implies \mathbf{x_2 = 2}$.
- Row 1: $x_1 + 2(2) + 3 = 8 \implies x_1 + 7 = 8 \implies \mathbf{x_1 = 1}$.

Solution: $(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = (1, 2, 3)$.

4. Free Variables & Infinite Solutions

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 2 & 4 & 1 & 11 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 3 & 3 \end{array} \right]$$

- Row 2: $3x_3 = 3 \implies \mathbf{x}_3 = 1$.
- Row 1: $x_1 + 2x_2 - 1 = 4 \implies x_1 = 5 - 2x_2$.

Here x_2 is a **free variable** ($x_2 = t$). General Parametric Solution: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 - 2t \\ t \\ 1 \end{bmatrix}$ for any $t \in \mathbb{R}$.

5. Data Science Application: Least Squares Normal Equations Using Cramer's Rule on $\begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 10 \\ 9 \end{bmatrix}$:

$$\det(A) = 4(3) - 2(2) = 12 - 4 = 8$$

$$w_0 = \frac{\begin{vmatrix} 10 & 2 \\ 9 & 3 \end{vmatrix}}{8} = \frac{30 - 18}{8} = \frac{12}{8} = \mathbf{1.5}$$

$$w_1 = \frac{\begin{vmatrix} 4 & 10 \\ 2 & 9 \end{vmatrix}}{8} = \frac{36 - 20}{8} = \frac{16}{8} = \mathbf{2.0}$$

Parameter Vector: $\mathbf{w} = \begin{bmatrix} 1.5 \\ 2.0 \end{bmatrix}$.

Week 3: Vector Spaces, Subspaces & Linear Independence

Detailed Solutions

1. Subspace Test for $W = \{[x, y]^T \mid 2x - 3y = 0\}$

- 1. Zero Vector:** For $\mathbf{0} = [0, 0]^T$, $2(0) - 3(0) = 0$. $\mathbf{0} \in W$.
- 2. Closure under Addition:** Let $\mathbf{u} = [x_1, y_1]^T, \mathbf{v} = [x_2, y_2]^T \in W$. Then $2x_1 - 3y_1 = 0$ and $2x_2 - 3y_2 = 0$. For $\mathbf{u} + \mathbf{v} = [x_1 + x_2, y_1 + y_2]^T$:

$$2(x_1 + x_2) - 3(y_1 + y_2) = (2x_1 - 3y_1) + (2x_2 - 3y_2) = 0 + 0 = 0 \implies \mathbf{u} + \mathbf{v} \in W$$

- 3. Closure under Scalar Multiplication:** For any scalar $c \in \mathbb{R}$:

$$2(cx_1) - 3(cy_1) = c(2x_1 - 3y_1) = c(0) = 0 \implies c\mathbf{u} \in W$$

Conclusion: W satisfies all 3 criteria and is a **valid subspace of $\mathbb{R}^2$** .

2. Non-Subspace Counterexample For $W = \{[x, y]^T \mid x + y = 1\}$:

- **Zero vector test:** $\mathbf{0} = [0, 0]^T \implies 0 + 0 = 0 \neq 1$. $\mathbf{0} \notin W$.

Failing the zero vector test immediately proves W is **NOT a vector subspace**.

3. Testing Linear Independence (3×3) Form matrix $A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3] = \begin{bmatrix} 1 & 0 & 5 \\ 2 & 1 & 6 \\ 3 & 4 & 0 \end{bmatrix}$. Compute

$\det(A)$ expanding along row 1:

$$\det(A) = 1(1 \cdot 0 - 4 \cdot 6) - 0 + 5(2 \cdot 4 - 3 \cdot 1) = 1(-24) + 5(8 - 3) = -24 + 25 = \mathbf{1}$$

Since $\det(A) = 1 \neq 0$, the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is **Linearly Independent**.

4. Linear Dependence Condition Vectors are linearly dependent iff $\det([\mathbf{u} \ \mathbf{v}]) = 0$:

$$\begin{vmatrix} 1 & 3 \\ 2 & k \end{vmatrix} = 1(k) - 3(2) = k - 6 = 0 \implies \mathbf{k} = 6$$

5. Data Science Application: Feature Multicollinearity Since 1 sq ft = 0.092903 sq m, $x_2 = 0.092903x_1$. The linear combination $0.092903 \cdot x_1 - 1 \cdot x_2 + 0 \cdot x_3 = 0$ holds with non-zero coefficients. Thus the feature set $\{x_1, x_2, x_3\}$ is **linearly dependent**. In linear regression, this exact linear dependence leads to a singular matrix $X^T X$ ($\det(X^T X) = 0$), making $(X^T X)^{-1}$ undefined. One of the redundant size features must be dropped.

Week 4: Basis, Dimension & Matrix Rank

Detailed Solutions

1. Basis Verification & Coordinate Representation Form matrix $M = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. $\det(M) = 1(-1) - 1(1) = -2 \neq 0$. Since $\det(M) \neq 0$, $\mathcal{B}$ consists of 2 linearly independent vectors in $\mathbb{R}^2$. Therefore, $\mathcal{B}$ is a **valid basis for $\mathbb{R}^2$** .

To find coordinates c_1, c_2 for $\mathbf{v} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$:

$$\begin{cases} c_1 + c_2 = 5 \\ c_1 - c_2 = 1 \end{cases} \implies 2c_1 = 6 \implies \mathbf{c_1 = 3, c_2 = 2}$$

Vector $\mathbf{v} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

2. Finding a Basis for Subspace W From $x + 2y - z = 0 \implies z = x + 2y$. Any vector in W can be written as:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ x + 2y \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

Basis for W : $\mathcal{B}_W = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$. Since there are 2 basis vectors, $\dim(\mathbf{W}) = 2$.

3. Matrix Rank Calculation

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 3 & 11 \\ 3 & 6 & 4 & 15 \end{bmatrix} \xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1}]{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The REF has 2 non-zero rows (pivots in column 1 and column 3). Therefore, $\text{rank}(\mathbf{A}) = 2$.

4. Full Rank vs Rank Deficient

- **Maximum Possible Rank:** $\min(100, 5) = 5$.
- **Interpretation if $\text{rank}(X) = 3$:** The dataset is **rank deficient**. Out of 5 feature columns, only 3 are linearly independent; 2 features are redundant linear combinations of the others.

5. Data Science Application: Collinear Feature Removal Form matrix $F = [\mathbf{f}_1 \ \mathbf{f}_2 \ \mathbf{f}_3] = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 3 & 6 & 5 \end{bmatrix}$.

Row operations:

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 3 & 6 & 5 \end{bmatrix} \xrightarrow[\substack{R_3 \rightarrow R_3 - 3R_1}]{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 5 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 5R_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

- **Matrix Rank:** $\text{rank}(\mathbf{F}) = 2$.
- **Redundant Feature:** $\mathbf{f}_2 = 2\mathbf{f}_1$ is completely redundant.
- **Minimal Basis:** $\{\mathbf{f}_1, \mathbf{f}_3\} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix} \right\}$.

Week 5: Null Space, Nullity & Linear Mappings

Detailed Solutions

1. Null Space Calculation Solve $A\mathbf{x} = \mathbf{0}$ for $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 5 \end{bmatrix}$:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 2 & 4 & 5 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 0 & 7 & 0 \end{array} \right]$$

- Row 2: $7x_3 = 0 \implies \mathbf{x}_3 = \mathbf{0}$.
- Row 1: $x_1 + 2x_2 - 0 = 0 \implies x_1 = -2x_2$.

Here $x_2 = t$ is a free variable.

$$\mathbf{x} = \begin{bmatrix} -2t \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

- **Basis for Null(A):** $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}$.
- **Nullity:** $\text{nullity}(A) = \dim(\text{Null}(A)) = 1$.

2. Testing Linearity of Mappings

- **(a)** $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x - y \\ x + 3y \end{bmatrix}$: Notice that $T(\mathbf{x}) = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$. Since T is induced by matrix multiplication, it is a **valid Linear Transformation**.
- **(b)** $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x^2 \\ y \end{bmatrix}$: Test scalar multiplication for $c = 2$, $\mathbf{v} = [1, 1]^T$:

$$T(2\mathbf{v}) = T\left(\begin{bmatrix} 2 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2^2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$2T(\mathbf{v}) = 2T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = 2\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Since $\begin{bmatrix} 4 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, T fails scalar homogeneity and is **NOT a Linear Transformation**.

3. Matrix-Induced Transformation

$$T\left(\begin{bmatrix} 3 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ -2 & 3 \\ 5 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1(3) + 0(-1) \\ -2(3) + 3(-1) \\ 5(3) + 4(-1) \end{bmatrix} = \begin{bmatrix} 3 \\ -9 \\ 11 \end{bmatrix}$$

4. Homogenous System Solutions For matrix $A \in \mathbb{R}^{3 \times 5}$:

- Domain dimension $n = 5$ (number of columns).
- By the Rank-Nullity Theorem: $\text{nullity}(A) = n - \text{rank}(A) = 5 - 3 = 2$.
- Since $\text{nullity}(A) = 2 > 0$, there exist **infinitely many non-trivial solutions** to $Ax = 0$.

5. Data Science Application: Linear Compression Null Space Matrix $W \in \mathbb{R}^{10 \times 50}$ maps inputs $\mathbf{x} \in \mathbb{R}^{50}$ ($n = 50$) to outputs $\mathbf{y} \in \mathbb{R}^{10}$. The maximum possible rank of W is $\min(10, 50) = 10$.

$$\text{nullity}(W) = n - \text{rank}(W) = 50 - \text{rank}(W) \geq 50 - 10 = 40$$

At least **40** dimensions of input variation are mapped to zero in the null space, representing the minimum lost information during linear compression.

Week 6: Matrix Representation, Kernel, Image & Rank-Nullity

Detailed Solutions

1. Matrix Representation Construction

$$T(\mathbf{e}_1) = T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3(1) + 0 \\ 2(1) - 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$T(\mathbf{e}_2) = T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 + 1 \\ 0 - 4(1) \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

$$A = [T]_{\mathcal{E}} = \begin{bmatrix} 3 & 1 \\ 2 & -4 \end{bmatrix}$$

2. Non-Standard Basis Matrix Representation Basis $\mathcal{B} = \left\{ \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$.

- Apply T to $\mathbf{v}_1$: $T(\mathbf{v}_1) = \begin{bmatrix} 1 + 1 \\ 2(1) - 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Express in basis $\mathcal{B}$: $c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \implies c_1 = 1, c_2 = 1 \implies [T(\mathbf{v}_1)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

- Apply T to $\mathbf{v}_2$: $T(\mathbf{v}_2) = \begin{bmatrix} 1 + 0 \\ 2(1) - 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Express in basis $\mathcal{B}$: $c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \implies c_1 = 2, c_2 = -1 \implies [T(\mathbf{v}_2)]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Matrix

Representation: $[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$.

3. Bases for Kernel and Image Row reduce $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{\text{REF}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$.

- **Pivots** are in Column 1 and Column 2.
- **(a) Basis for $\text{im}(A)$:** Take original columns 1 and 2 of A :

$$\mathcal{B}_{\text{im}} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \right\}, \quad \text{rank}(A) = 2$$

- **(b) Basis for $\ker(A)$:** Solve REF $A\mathbf{x} = \mathbf{0}$:
- Row 2: $-x_2 - 2x_3 = 0 \implies x_2 = -2x_3$.
- Row 1: $x_1 + 2(-2x_3) + 3x_3 = 0 \implies x_1 = x_3$.

Set $x_3 = t$: $\mathbf{x} = t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$.

$$\mathcal{B}_{\ker} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\}, \quad \text{nullity}(A) = 1$$

4. Rank-Nullity Verification Domain $V = \mathbb{R}^3$, so $\dim(V) = 3$.

$$\text{nullity}(A) + \text{rank}(A) = 1 + 2 = \mathbf{3} = \dim(\mathbb{R}^3) \quad \checkmark$$

5. Data Science Application: Linear Autoencoder Information Loss Given $W \in \mathbb{R}^{20 \times 100}$ and $\text{rank}(W) = 18$:

- **(a) Code Space Dimension:** $\dim(\text{im}(W)) = \text{rank}(W) = \mathbf{18}$.
- **(b) Lost Feature Dimensions:** By Rank-Nullity Theorem:

$$\dim(\ker(W)) = 100 - \text{rank}(W) = 100 - 18 = \mathbf{82} \text{ lost feature dimensions}$$

Week 7: Matrix Similarity, Affine Spaces, Norms & Inner Products

Detailed Solutions

1. Matrix Similarity Check

- **(a) Determinants:** $\det(A) = 1(3) - 0 = \mathbf{3}$, $\det(B) = 3(1) - 0 = \mathbf{3}$.
- **(b) Traces:** $\text{tr}(A) = 1 + 3 = \mathbf{4}$, $\text{tr}(B) = 3 + 1 = \mathbf{4}$.
- **(c) Similarity:** The eigenvalues of upper triangular A are $\lambda_1 = 1, \lambda_2 = 3$. The eigenvalues of lower triangular B are $\lambda_1 = 3, \lambda_2 = 1$. Since both have distinct eigenvalues $\{1, 3\}$, both matrices are diagonalizable to $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$. Therefore, A and B are **SIMILAR** ($A \sim B$).

2. Weighted Inner Product

- **(a) Inner Product:**

$$\left\langle \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \end{bmatrix} \right\rangle_Q = 2(1)(3) + 5(2)(-1) = 6 - 10 = \mathbf{-4}$$

- **(b) Induced Norm:**

$$\left\| \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\|_Q = \sqrt{2(1)^2 + 5(2)^2} = \sqrt{2 + 20} = \sqrt{\mathbf{22}}$$

3. Cosine Similarity

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = (1)(0) + (2)(1) + (0)(1) + (2)(1) = 0 + 2 + 0 + 2 = 4$$

$$\|\mathbf{d}_1\| = \sqrt{1^2 + 2^2 + 0^2 + 2^2} = \sqrt{9} = 3$$

$$\|\mathbf{d}_2\| = \sqrt{0^2 + 1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\cos(\theta) = \frac{4}{3\sqrt{3}} \approx \mathbf{0.7698}$$

4. Cauchy-Schwarz Verification

$$\mathbf{u} \cdot \mathbf{v} = 3(1) + 4(2) = 11 \implies |\mathbf{u} \cdot \mathbf{v}| = 11$$

$$\|\mathbf{u}\|\|\mathbf{v}\| = \left(\sqrt{3^2 + 4^2}\right) \left(\sqrt{1^2 + 2^2}\right) = 5\sqrt{5} \approx \mathbf{11.1803}$$

Since $11 \leq 11.1803$, the **Cauchy-Schwarz inequality** $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|$ holds true.

5. Data Science Application: L_1 vs L_2 Norm Regularization

- (a) L_1 Norm (Lasso):

$$\|\mathbf{w}\|_1 = |3| + |-4| + |0| + |12| = 3 + 4 + 0 + 12 = \mathbf{19}$$

- (b) L_2 Norm (Ridge):

$$\|\mathbf{w}\|_2 = \sqrt{3^2 + (-4)^2 + 0^2 + 12^2} = \sqrt{9 + 16 + 0 + 144} = \sqrt{169} = \mathbf{13}$$

Week 8: Orthogonality, Gram-Schmidt Process & Projections**Detailed Solutions****1. Projection onto a Vector**

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{v} \cdot \mathbf{u}}{\|\mathbf{u}\|^2} \mathbf{u} = \frac{4(1) + 3(2)}{1^2 + 2^2} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{10}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

2. Gram-Schmidt Process (2×2)

$$1. \text{ Step 1: } \mathbf{u}_1 = \mathbf{v}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}. \|\mathbf{u}_1\| = \sqrt{3^2 + 4^2} = 5.$$

$$\mathbf{q}_1 = \frac{\mathbf{u}_1}{5} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

$$2. \text{ Step 2: } \mathbf{u}_2 = \mathbf{v}_2 - (\mathbf{v}_2 \cdot \mathbf{q}_1)\mathbf{q}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \left(1 \cdot \frac{3}{5} + 0\right) \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \begin{bmatrix} 1 - 9/25 \\ -12/25 \end{bmatrix} = \begin{bmatrix} 16/25 \\ -12/25 \end{bmatrix}. \|\mathbf{u}_2\| = \sqrt{(16/25)^2 + (-12/25)^2} = \frac{20}{25} = \frac{4}{5}.$$

$$\mathbf{q}_2 = \frac{\mathbf{u}_2}{4/5} = \begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix}$$

Orthonormal Basis: $\left\{ \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}, \begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix} \right\}.$

3. Gram-Schmidt Process (3×3)

$$\bullet \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \|\mathbf{u}_1\| = \sqrt{2} \Rightarrow \mathbf{q}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}.$$

$$\bullet \mathbf{u}_2 = \mathbf{v}_2 - (\mathbf{v}_2 \cdot \mathbf{q}_1)\mathbf{q}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}.$$

$$\|\mathbf{u}_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}} \Rightarrow \mathbf{q}_2 = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}.$$

$$\bullet \text{Repeating for } \mathbf{u}_3 \Rightarrow \mathbf{q}_3 = \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ -1/\sqrt{3} \end{bmatrix}.$$

4. Orthogonal Matrix Properties

- (a) $Q^T Q$ Computation:

$$Q^T Q = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & \sin^2 \theta + \cos^2 \theta \end{bmatrix} = \mathbf{I}$$

- (b) Norm Preservation:

$$\|Q\mathbf{x}\|^2 = (Q\mathbf{x})^T(Q\mathbf{x}) = \mathbf{x}^T(Q^T Q)\mathbf{x} = \mathbf{x}^T \mathbf{I} \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|^2 \Rightarrow \|Q\mathbf{x}\| = \|\mathbf{x}\|$$

5. Data Science Application: Projection Matrix Properties Given $P = X(X^T X)^{-1} X^T$:

- Symmetry ($P^T = P$):

$$P^T = (X(X^T X)^{-1} X^T)^T = (X^T)^T ((X^T X)^{-1})^T X^T = X(X^T X)^{-1} X^T = \mathbf{P}$$

- Idempotence ($P^2 = P$):

$$P^2 = PP = X(X^T X)^{-1} \underbrace{(X^T X)(X^T X)^{-1}}_{\mathbf{I}} X^T = X(X^T X)^{-1} X^T = \mathbf{P}$$

Week 9: Multivariable Functions, Partial Derivatives & Gradient

Detailed Solutions

1. First-Order Partial Derivatives For $f(x, y) = x^2 y^3 + e^{2x} \sin(y)$:

$$\frac{\partial f}{\partial x} = 2xy^3 + 2e^{2x} \sin(y)$$

$$\frac{\partial f}{\partial y} = 3x^2 y^2 + e^{2x} \cos(y)$$

2. Gradient Vector Evaluation For $f(x, y) = 4x^2 y - 3y^2 + 5x$:

$$\frac{\partial f}{\partial x} = 8xy + 5 \Rightarrow f_x(1, 2) = 8(1)(2) + 5 = 16 + 5 = 21$$

$$\frac{\partial f}{\partial y} = 4x^2 - 6y \Rightarrow f_y(1, 2) = 4(1)^2 - 6(2) = 4 - 12 = -8$$

$$\nabla f(1, 2) = \begin{bmatrix} 21 \\ -8 \end{bmatrix}$$

3. Directional Derivative 1. Gradient $\nabla f(x, y) = \begin{bmatrix} 2x + 3y \\ 3x + 2y \end{bmatrix}$. At $(2, 1)$: $\nabla f(2, 1) = \begin{bmatrix} 2(2) + 3(1) \\ 3(2) + 2(1) \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$. 2. Normalize direction vector $\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \implies \mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$. 3. Directional derivative:

$$D_{\mathbf{u}}f(2, 1) = \nabla f(2, 1) \cdot \mathbf{u} = \begin{bmatrix} 7 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \frac{21 + 32}{5} = \frac{53}{5} = 10.6$$

4. Steepest Ascent Direction For $f(x, y) = x^3y - 2xy^2$:

- $f_x = 3x^2y - 2y^2 \implies f_x(1, -1) = 3(1)(-1) - 2(1) = -5$.
- $f_y = x^3 - 4xy \implies f_y(1, -1) = 1 - 4(1)(-1) = 1 + 4 = 5$.

Gradient: $\nabla f(1, -1) = \begin{bmatrix} -5 \\ 5 \end{bmatrix}$.

- **Unit Direction of Steepest Ascent:** $\mathbf{u} = \frac{\nabla f}{\|\nabla f\|} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$.
- **Maximum Rate of Increase:** $\|\nabla f(1, -1)\| = \sqrt{(-5)^2 + 5^2} = \sqrt{50} = 5\sqrt{2} \approx 7.071$.

5. Data Science Application: MSE Loss Gradient For $L(m, b) = (mx_i + b - y_i)^2$: Using the chain rule:

$$\frac{\partial L}{\partial m} = 2(mx_i + b - y_i) \cdot x_i = 2x_i(mx_i + b - y_i)$$

$$\frac{\partial L}{\partial b} = 2(mx_i + b - y_i) \cdot 1 = 2(mx_i + b - y_i)$$

$$\nabla L(m, b) = \begin{bmatrix} 2x_i(mx_i + b - y_i) \\ 2(mx_i + b - y_i) \end{bmatrix}$$

Week 10: Tangent Planes, Steepest Descent & Critical Points

Detailed Solutions

1. Tangent Plane Equation For $z = x^2 + 3y^2 - 4x$ at $(x_0, y_0) = (2, 1)$:

- $z_0 = f(2, 1) = 2^2 + 3(1)^2 - 4(2) = 4 + 3 - 8 = -1$.
- $f_x = 2x - 4 \implies f_x(2, 1) = 2(2) - 4 = 0$.
- $f_y = 6y \implies f_y(2, 1) = 6(1) = 6$.

Tangent Plane:

$$z - (-1) = 0(x - 2) + 6(y - 1) \implies z + 1 = 6y - 6 \implies \mathbf{z = 6y - 7}$$

2. Gradient Descent Iteration Step

- **(a) Gradient:** $\nabla L(w_1, w_2) = \begin{bmatrix} 2w_1 \\ 8w_2 \end{bmatrix}$.

- **(b) Update Step:** At $\mathbf{w}^{(0)} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$: $\nabla L(3, 2) = \begin{bmatrix} 6 \\ 16 \end{bmatrix}$.

$$\mathbf{w}^{(1)} = \mathbf{w}^{(0)} - \alpha \nabla L = \begin{bmatrix} 3 \\ 2 \end{bmatrix} - 0.1 \begin{bmatrix} 6 \\ 16 \end{bmatrix} = \begin{bmatrix} 3 - 0.6 \\ 2 - 1.6 \end{bmatrix} = \begin{bmatrix} 2.4 \\ 0.4 \end{bmatrix}$$

3. Finding Critical Points (2×2) Set first partial derivatives of $f(x, y) = x^3 + y^3 - 3xy$ to zero:

1. $f_x = 3x^2 - 3y = 0 \implies y = x^2$.

2. $f_y = 3y^2 - 3x = 0 \implies x = y^2$.

Substitute $y = x^2$ into $x = y^2$:

$$x = (x^2)^2 = x^4 \implies x^4 - x = 0 \implies x(x^3 - 1) = 0$$

- If $x = 0 \implies y = 0^2 = 0 \implies$ Critical point $(0, 0)$.
- If $x = 1 \implies y = 1^2 = 1 \implies$ Critical point $(1, 1)$.

4. Linear Approximation For $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$: $f(3, 4) = 5$.

- $f_x = \frac{x}{\sqrt{x^2 + y^2}} \implies f_x(3, 4) = \frac{3}{5} = 0.6$.
- $f_y = \frac{y}{\sqrt{x^2 + y^2}} \implies f_y(3, 4) = \frac{4}{5} = 0.8$.

Linear approximation function: $L(x, y) = 5 + 0.6(x - 3) + 0.8(y - 4)$. For $(3.06, 3.92)$:

$$L(3.06, 3.92) = 5 + 0.6(0.06) + 0.8(-0.08) = 5 + 0.036 - 0.064 = \mathbf{4.972}$$

5. Data Science Application: Least Squares Gradient Descent Update For $L(w) = (2w - 6)^2$:

$$\frac{dL}{dw} = 2(2w - 6) \cdot 2 = 8w - 24$$

- **Step 1:** At $w^{(0)} = 0$, $\frac{dL}{dw} = -24 \implies w^{(1)} = 0 - 0.05(-24) = \mathbf{1.2}$.
- **Step 2:** At $w^{(1)} = 1.2$, $\frac{dL}{dw} = 8(1.2) - 24 = -14.4 \implies w^{(2)} = 1.2 - 0.05(-14.4) = 1.2 + 0.72 = \mathbf{1.92}$.

The weight sequence $0 \rightarrow 1.2 \rightarrow 1.92$ rapidly converges toward the optimal minimum $w^* = 3$.

Week 11: Hessian Matrix, Multivariable Extrema & Differentiability

Detailed Solutions

1. Hessian Matrix Computation For $f(x, y) = x^3 - 3xy^2 + y^4$:

- First partials: $f_x = 3x^2 - 3y^2$, $f_y = -6xy + 4y^3$.
- Second partials: $f_{xx} = 6x$, $f_{xy} = -6y$, $f_{yy} = -6x + 12y^2$.

$$H(f)(x, y) = \begin{bmatrix} 6x & -6y \\ -6y & -6x + 12y^2 \end{bmatrix}$$

At point $(1, 2)$:

$$H(f)(1, 2) = \begin{bmatrix} 6(1) & -6(2) \\ -6(2) & -6(1) + 12(2^2) \end{bmatrix} = \begin{bmatrix} 6 & -12 \\ -12 & 42 \end{bmatrix}$$

2. Classifying 2D Critical Points For $f(x, y) = x^3 + y^3 - 3xy$, second partials are $f_{xx} = 6x$, $f_{xy} = -3$, $f_{yy} = 6y$. Hessian Discriminant: $D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 = (6x)(6y) - (-3)^2 = 36xy - 9$.

- **At (0, 0):** $D(0, 0) = 36(0) - 9 = -9 < 0$. Therefore, (0, 0) is a **Saddle Point**.
- **At (1, 1):** $D(1, 1) = 36(1)(1) - 9 = 27 > 0$, and $f_{xx}(1, 1) = 6 > 0$. Therefore, (1, 1) is a **Local Minimum**.

3. Saddle Point Identification For $f(x, y) = x^2 - y^2$ at critical point (0, 0):

$$H(f)(0, 0) = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

Determinant $D = 2(-2) - 0 = -4 < 0$. Since $D < 0$, the Hessian is indefinite with eigenvalues $\lambda_1 = 2 > 0$ and $\lambda_2 = -2 < 0$. Thus (0, 0) is a **Saddle Point**.

4. Classifying 3D Critical Points For $f(x, y, z) = x^2 + 2y^2 + 3z^2 - 4x + 8y - 12z$:

- First partials: $f_x = 2x - 4 = 0 \implies x = 2$, $f_y = 4y + 8 = 0 \implies y = -2$, $f_z = 6z - 12 = 0 \implies z = 2$.

Critical point: **(2, -2, 2)**.

- Hessian matrix:

$$H(f)(2, -2, 2) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

All diagonal entries (eigenvalues) are strictly positive (2, 4, 6 > 0), meaning $H(f)$ is positive-definite. Therefore, (2, -2, 2) is a **Local Minimum**.

5. Data Science Application: Convexity of Mean Squared Error For $L(w) = \frac{1}{N} \sum_{i=1}^N (wx_i - y_i)^2$:

- First derivative: $\frac{dL}{dw} = \frac{2}{N} \sum_{i=1}^N x_i (wx_i - y_i)$.
- Second derivative: $\frac{d^2L}{dw^2} = \frac{2}{N} \sum_{i=1}^N x_i^2$.

Since $x_i^2 \geq 0$ for all real sample features x_i , the second derivative $\frac{d^2L}{dw^2} \geq 0$ for all w . Because the second derivative is non-negative everywhere, $L(w)$ is **strictly convex**, ensuring that any local minimum found by Gradient Descent is the **unique global minimum**.

Week 12: Eigenvalues, Eigenvectors, Diagonalization & SVD

Detailed Solutions

1. Eigenvalues & Eigenvectors (2×2)

$$\det(A - \lambda I) = \begin{vmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{vmatrix} = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0$$

- Eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 2$.

- **Eigenvector for $\lambda_1 = 5$:** $(A - 5I)\mathbf{v}_1 = \mathbf{0} \implies \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0} \implies x = y \implies \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

- **Eigenvector for $\lambda_2 = 2$:** $(A - 2I)\mathbf{v}_2 = \mathbf{0} \implies \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0} \implies 2x + y = 0 \implies y = -2x \implies \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.

2. Matrix Diagonalization & Matrix Powers

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}, \quad P^{-1} = \frac{1}{-3} \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

To compute $A^4 = PD^4P^{-1}$:

$$\begin{aligned} D^4 &= \begin{bmatrix} 5^4 & 0 \\ 0 & 2^4 \end{bmatrix} = \begin{bmatrix} 625 & 0 \\ 0 & 16 \end{bmatrix} \\ A^4 &= \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 625 & 0 \\ 0 & 16 \end{bmatrix} \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 625 & 16 \\ 625 & -32 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \\ A^4 &= \frac{1}{3} \begin{bmatrix} 1266 & 609 \\ 1218 & 657 \end{bmatrix} = \begin{bmatrix} 422 & 203 \\ 406 & 219 \end{bmatrix} \end{aligned}$$

3. Symmetric Matrix Spectral Decomposition For $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$: $\det(A - \lambda I) = (3 - \lambda)^2 - 1 = 0 \implies (3 - \lambda) = \pm 1 \implies \lambda_1 = 4, \lambda_2 = 2$.

- For $\lambda_1 = 4$: Normalized unit eigenvector $\mathbf{q}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$.
- For $\lambda_2 = 2$: Normalized unit eigenvector $\mathbf{q}_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$.

Orthogonal Matrix $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ satisfies $Q^T Q = I$ and $A = Q D Q^T$.

4. Singular Values Calculation For $A = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$:

$$A^T A = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$$

Eigenvalues of $A^T A$ are $\lambda_1 = 9, \lambda_2 = 4$. Singular Values: $\sigma_1 = \sqrt{9} = 3, \sigma_2 = \sqrt{4} = 2$.

5. Comprehensive Course Review: PCA via SVD

- **(a) Why Principal Components are Eigenvectors of C :** The covariance matrix $C = \frac{1}{N} X^T X$ measures feature variances and co-variances. The eigenvectors of C define orthogonal axes along which feature variance is maximized, decorrelating features.
- **(b) Percentage of Variance Explained:**

$$\text{Total Variance} = \lambda_1 + \lambda_2 + \lambda_3 = 80 + 15 + 5 = 100$$

$$\text{Explained Variance by PC1 \& PC2} = \frac{\lambda_1 + \lambda_2}{\text{Total}} = \frac{80 + 15}{100} = \frac{95}{100} = 95\%$$